Nicholas Eisenberg, Ph.D.

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Experience

Senior Data Scientist *July 2022 - Present*
Nevada National Security Site

- Built and managed a MySQL database of seismic sensor data using SQLAlchemy, optimizing data retrieval for analysis purposes
- Acted as the lead investigator for project involving the organization and feature engineering of a dataset consisting of 10,000+ shots from the Los Alamos National Lab linear accelerator
- Constructed a predictive model using Keras to forecast future performance of an X-ray machine with accuracy of %90, resulting in improved maintenance planning and operational efficiency
- Developed interactive dashboards using Streamlit and Panel to visually present complex data and provide actionable insights to fellow scientists and stakeholders
- Applied Scikit-learn and Scikit-image to characterize the spot of an X-ray machine, thus estimating the blur kernel for deblurring X-ray images, enhancing image quality, and aiding in accurate analysis
- Acted as lead maintainer of a company-wide internal Python library hosted on our internal GitLab, ensuring code quality, version control, and collaboration amongst team members
- Utilized Neovim, Tmux and Vim-Slime along with SSH and WinSCP to run code on our interal Linux cluster

Graduate Teaching Assistant *Janurary 2016 - June 2022*
Auburn University and University of Nevada Las Vegas

- Instructed courses in College Algebra and Pre-Calculus
- Led weekly recitation sections for Mathematical Statistics, Real Analysis and Calculus I, II and III
- Managed Auburn University's Math tutoring center of 14 tutors

Technical Skills

Programming Languages: Python, SQL, R, Lua, Bash, LaTeX

Libraries: scikit-learn, tensorflow, keras, pytorch, scipy, plotly, panel, pyarrow, pymysql, sqlalchemy

Developer Tools: Git, Unix/Linux, Neovim, Tmux, SSH

Publications

1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). https://doi.org/10.1007/s40072-022-00258-6

2: Eisenberg, N., Chen, L. Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term. Submitted Janurary 2023 to Journal of Theoretical Probability.
Preprint available at https://arxiv.org/pdf/2209.04771v1.pdf

Education

Auburn University, Ph.D. *June 2022*
Research Area: Probability (Stochastic Processes and Partial Differential Equtions)
Advisor: Le Chen

University of Nevada Las Vegas, B.S./M.S. *December 2015 / Janurary 2018*
Major: Applied Mathematics